# INFO 4220 – Networks II: Market Design
# Spring 2025 – Problem Set 2

**Solution**

## Problem 1 (10 points)

Consider the following one-to-one matching problem with 2 musicians and 2 singers. The preferences are as follows:

| $P_{m_1}$ | $P_{m_2}$ |
|---|---|
| $s_1$ | $s_2$ |
| $s_2$ | $s_1$ |

| $P_{s_1}$ | $P_{s_2}$ |
|---|---|
| $m_1$ | $m_2$ |
| $m_2$ | $m_1$ |

Assume the deferred acceptance algorithm with singers proposing. Is it a dominant strategy for musicians to report truthfully their preferences? Why (do not use a theorem to explain? (hint: consider carefully the definition of strategyproof).

**Solution:**
A dominant strategy is a strategy that is the best to use for whatever strategies the other players use.
In the case, being truthful is a dominant strategy for musicians if:

- Each musician being truthful is the best strategy if singers are being truthful
- Each musician being truthful is the best strategy if singers are not being truthful

Suppose that singers, instead of submitting their true preferences submit:

| $P_{s_1'}$ | $P_{s_2'}$ |
|---|---|
| $m_2$ | $m_1$ |
| $m_1$ | $m_2$ |

Now, suppose that $m_2$ is being truthful and let's see what happens with $m_1$:

- If $m_1$ is truthful, he ends up matched with $s_2$
- If $m_1$ is not truthful and submits $P'm_1 = s_1$, $m_1$ he ends up matched with $s_1$, which is an outcome he prefers.

Thus, it is not always a dominant strategy for musicians to be truthful. Note that while in this case it doesn't make sense to lie if singers submit their true preferences, the definition of dominant strategy requires that you consider ALL possible strategies singers may use (even the one that are not optimal for them).

## Problem 2 (20 points, 5 pts each)

Consider a medical match problem with three hospitals: $h_1, h_2, h_3$ and four doctors $d_1, d_2, d_3, d_4$. The capacities of the hospitals are: $q_{h_1} = 2, q_{h_2} = 1, q_{h_3} = 1$. The preferences of doctors and hospitals are as follows:

| $P_{d_1}$ | $P_{d_2}$ | $P_{d_3}$ | $P_{d_4}$ |
|---|---|---|---|
| $h_3$ | $h_2$ | $h_1$ | $h_1$ |
| $h_1$ | $h_1$ | $h_3$ | $h_2$ |
| $h_2$ | $h_3$ | $h_2$ | $h_3$ |

| $P_{h_1}$ | $P_{h_2}$ | $P_{h_3}$ |
|---|---|---|
| $d_1$ | $d_1$ | $d_3$ |
| $d_2$ | $d_2$ | $d_1$ |
| $d_3$ | $d_3$ | $d_2$ |
| $d_4$ | $d_4$ | $d_4$ |

a) Find the doctor-optimal matching

**Solution:**

Step 1
$d_3, d_4$ propose to $h_1$
$d_2$ proposes $h_2$
$d_3$ proposes $h_3$

There are no rejections and the algorithm ends.

b) Find the hospital-optimal matching

**Solution:**

Step 1:
$h_1$ proposes to $d_1, d_2$
$h_2$ proposes to $d_1$
$h_3$ proposes to $d_3$
$d_1$ rejects $h_2$

Step 2:
$h_2$ proposes to $d_2$
$d_2$ rejects $h_1$

Step 3:
$h_1$ proposes to $d_3$
$d_3$ rejects $h_3$

Step 4:
$h_3$ proposes to $d_1$
$d_1$ rejects $h_1$

Step 5:
$h_1$ proposes to $d_4$
NO rejections, algorithm ends $\mu(h_1) = d_3, d_4$ ; $\mu(h_2) = d_2; \mu(h_3) = d_1$

c) Consider the following matching $\mu(h_1) = d_2, d_4; \mu(h_2) = d_1, \mu(h_3) = d_3$. If we use deferred acceptance with hospitals proposing, is there a preference list over doctors that hospitals have to submit to obtain this matching?

**Solution:**

Yes, if $h_1$ lists as its first two preferences $d_2, d_4$, and $h_2$ puts $d_1$ as its first preference, and $h_3$ lists $d_3$ as first references, the following will happen in D.A. w/ hospitals proposing:
$h_1$ will propose to $d_2, d_4$
$h_2$ will propose to $d_1$
$h_3$ will propose to $d_3$
As each doctor only received one offer, the algorithm ends with the matching we wanted.

d) True/False (explain): The deferred acceptance algorithm with hospitals proposing is strategyproof for hospitals, but it is not strategyproof for doctors.

**Solution:**
False. D.A. w/ hospitals proposing is not strategyproof for hospitals (see part c). It is also not strategyproof for doctors (i.e. that part of the statement was true)

## Problem 3 (20 points, 10 pts each)

In a medical residence matching program there are 3 hospitals and 6 doctors. Hospital 1 can admit up to 3 doctors. Hospitals 2 and 3 are smaller and can only admit 2 residents each. The true preferences of doctors over hospitals and hospitals over doctors are given by the following tables (assume hospitals preferences are responsive). Note that doctors $d_5$ and $d_6$ are a couple and their preferences are expressed as pairs of hospitals: The first preference of $d_5$ & $d_6$ is to both go to $h_1$, their second preference is to both go to $h_3$, and their third preference is to both go to $h_2$.

| $\boldsymbol{d_1}$ | $\boldsymbol{d_2}$ | $\boldsymbol{d_3}$ | $\boldsymbol{d_4}$ | $\boldsymbol{d_5}$ **&** $\boldsymbol{d_6}$ |
|---|---|---|---|---|
| $h_1$ | $h_1$ | $h_1$ | $h_1$ | $(h_1, h_1)$ |
| $h_3$ | $h_2$ | $h_3$ | $h_2$ | $(h_3, h_3)$ |
| $h_2$ | $h_3$ | $h_2$ | $h_3$ | $(h_2, h_2)$ |

| $\boldsymbol{h_1}$ | $\boldsymbol{h_2}$ | $\boldsymbol{h_3}$ |
|---|---|---|
| $d_5$ | $d_5$ | $d_6$ |
| $d_4$ | $d_3$ | $d_2$ |
| $d_1$ | $d_1$ | $d_3$ |

| | | |
|---|---|---|
| $d_2$ | $d_2$ | $d_1$ |
| $d_3$ | $d_4$ | $d_4$ |
| $d_6$ | $d_6$ | $d_5$ |

a) Use the following algorithm to match hospitals and doctors (15 pts):

**Solution:**

Step 1: Use the DA algorithm with doctors proposing excluding the couple ($d_5$ & $d_6$) from the problem
Step 2: Now manually match the couple to hospitals in order of their preferences (you must match them as a couple). When doing this manual matching, consider that the couple can displace doctors that are already matched as long as you leave hospital hiring them with a set of doctors that is preferred to the matchings they had obtained in step 1.
Step 3: Now manually match the doctor(s) displaced in step 2 to a hospital in order of their preferences. In this step unmatched doctors cannot displace doctors that have already been matched, so they can only be matched to hospitals that have a vacancy.

Step 1: We use DA without considering $d_5$ and $d_6$

- Step 1.1
  - $d_1$, $d_2$, $d_3$ and $d_4$ propose to $h_1$
  - $h_1$ rejects $d_3$
- Step 1.2
  - $D_3$ proposes to $h_3$.
  - END:: $\mu(h_1) = (d_1, d_2, d_4), \mu(h_3) = d_3$

Step 2: We try manually to match ($d_5$, $d_6$) to $h_1$. We can do this as long as we leave $h_1$ with a set of doctors it prefers. To compare sets of doctors, we need to use responsive preferences. We cannot compare ($d_5$, $d_6$, $d_4$) with ($d_1$, $d_2$, $d_4$) using responsive preferences. Thus, it is impossible to know if $h_1$ would be better off, and we cannot match ($d_5$,$d_6$) to $h_1$.
Next, we try to manually match ($d_5$,$d_6$) to $h_3$. In this case, we know that $(d_5, d_6) P_H^{\#}(d_3, \emptyset)$, so the couple can displace $d_3$ and now $\mu(h_3) = (d_5, d_6)$

Step 3: We match $d_3$ to $h_2$. The final matching is $\mu(h_1) = (d_1, d_2, d_4), \mu(h_3) = (d_5, d_6), \mu(h_2) = (d_3, \emptyset)$.

b) For the doctor(s) displaced in step 2. Can you find a hospital that would be willing to form a blocking pair with them? (Note that you must treat the couple as an indivisible couple: If you want to remove one doctor of the couple from a hospital, you must remove both). (5 pts)

**Solution:**

Only $d_3$ was displaced. This doctor prefers $h_1$ and $h_3$ to their current match. However, none of these would like to form a blocking pair because:

- $d_3$ is ranked lower than the doctors currently matched to $h_1$.
- While $h_3$ likes $d_3$ better than $d_5$, if $h_3$ changes $d_3$ for $d_5$, then $d_6$ will also leave and we know $(d_5, d_6)P_H^{\#}(d_3, \emptyset)$,